A gauging lemma for anomalies of holomorphic SVOAs: gauging an odd abelian normal subgroup with inflated anomaly

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(companion to `gauging_lemma.md`, §§G0–G11, and `NOTES_q8_anomaly.md`, §S6.3(d))

Abstract

Let $H = E \rtimes Q$ act on a holomorphic SVOA V, where E is abelian of odd order coprime to $|Q|$ and the anomaly of H is inflated from $\alpha \in \widetilde{SH}^3(BQ)$. We prove that for every Q-invariant discrete torsion ψ, the quotient Q acts on the orbifold $V/\!/_\psi E$, and that this action has anomaly α. Majorana, Gu–Wen and bosonic layers are all treated. The proof is categorical. Its VOA inputs are the algebra-object description of twisted modules (Kirillov; Huang–Kirillov–Lepowsky; Creutzig–Kanade–McRae; McRae), applied after adjoining fermion parity, together with Carnahan–Miyamoto regularity and quantum Galois theory. This is the lemma used in §S6 of the Q8-anomaly notes to show that the Q_8 action on the length-96 code SVOA is non-anomalous.

Status labels. **[P]** proved here; **[C]** cited, with theorem number checked against the source text unless stated otherwise; **[C°]** cited, number or attribution not re-checked. Section 9 collects them.

Authority. This document is the authoritative version of the gauging lemma. `gauging_lemma.md` (§§G0–G13) is the working record of its derivation and of the review rounds; where the two differ, this document is correct.

Conventions.

- SVOAs are of CFT type with a positive-definite invariant form.
- Groups act by even automorphisms, which commute with fermion parity $\sigma = (-1)^F$.
- For a group automorphism we write ${}^q e := qeq^{-1}$. For a 2-cochain ψ on E we put $\psi^q(a,b) := \psi({}^{q^{-1}}a, {}^{q^{-1}}b)$.
- For a finite group $G \subset \mathrm{Aut}(V)$, $\mathcal{F}(V,G) = \bigoplus_{g\in G} \mathcal{F}_g$ is the G-graded super-fusion category of g-twisted modules of Gaiotto–Johnson-Freyd [1, Thm. 4.2]. Its precise construction is recalled in §2.

1 Statement

Theorem 1.1 (Gauging lemma)**.** *Let V be a holomorphic SVOA and $H \subset \mathrm{Aut}(V)$ a finite solvable group. Suppose $H = E \rtimes Q$ with $E \trianglelefteq H$. Write $\pi : H \to Q$ for the projection and $s : Q \to H$ for the complement. Assume:*

(0) E is abelian of odd order and $\gcd(|E|,|Q|) = 1$;

(1) E is non-anomalous, so that the holomorphic orbifolds $V/\!/_\psi E$, $\psi \in H^2(E;U(1))$, exist;

(2) $\alpha(V,H) = \pi^*\alpha$ for some $\alpha \in \widetilde{SH}^3(BQ)$; necessarily $\alpha = s^*\alpha(V,H) = \alpha(V,Q)$;

(3) $[\psi] \in H^2(E; U(1))$ *is fixed by* Q*, acting through conjugation in* H*. By Lemma 3.4 this is equivalent to the intrinsic condition* $s(q) \cdot V//_\psi E \cong V//_\psi E$ *over* V^E *for all* q*;*

(4) the Majorana layer vanishes: $m(V, Q) = 0 \in H^1(Q; \mathbb{Z}_2)$*;*

(5) (P) $\sigma \notin H$*;*

(6) (R) *the even part* V_0 *is a regular VOA, and the twisted sectors of* $V' := V//_\psi E$ *other than* V^E *have positive conformal weights.*

Then:

(i) The action of H *on* V^E *extends to a group* $K \subset \mathrm{Aut}(V')$*, which is an extension* $1 \to \widehat{E} \to K \to Q \to 1$ *with* $\widehat{E} = \mathrm{Hom}(E, U(1))$ *the dual symmetry.* K *splits, any two complements of* $\widehat{E}$ *are conjugate by an element of* $\widehat{E}$*, and* $(V')^K = V^H$*.*

(ii) For every complement $Q' \subset K$ *of* $\widehat{E}$*, identified with* Q *via* $K \to Q$*,*

$$\alpha(V', Q') = \alpha \in \widetilde{SH}^3(BQ).$$

Remark 1.2 (Where the hypotheses enter)**.**

- *Odd order of* E*.* The parity homomorphism $E \to \mathbb{Z}_2$ is trivial (Lemma 3.1 and Lemma 4.2). The Majorana and Gu–Wen layers of E vanish, so discrete torsion is ordinary H^2. Lyndon–Hochschild–Serre gives $H^i(H; \mathbb{Z}_2) \cong H^i(Q; \mathbb{Z}_2)$.
- E *abelian.* The quantum symmetry is a group $\widehat{E}$ with $|\widehat{E}| = |E|$, and the Schur–Weyl pieces are invertible. Lemma 4.2 splits the right action into characters.
- *Coprimality.* Invariant cocycles (Lemma 3.2), $H^1(Q; \widehat{E}) = 0$, and Schur–Zassenhaus.
- $m = 0$. Simple objects of $\mathcal{F}(V, Q)$ have $\mathrm{End} = \mathbb{C}$.
- *(P) and (R).* These are used only to apply the algebra-object theory over a bosonic base (§2). If $\sigma \in H$, McRae's theorem applies to H directly.

Hypothesis (2) implies (1), because $\mathrm{res}_E \pi^* \alpha = 0$. Conversely, under (0), (1) implies (2) (Proposition 2.3).

Remark 1.3 (The application)**.** In §S6 of the notes, $V = V_{\Lambda(C_i)}$, $E = (C_i + U_b)/C_i \cong \mathbb{Z}_3^2$ and $Q = Q_8$ with the Gaiotto–Johnson-Freyd lift, so $|H| = 72$. All hypotheses hold:

- (1) and (2): S6.3(b) and S6.3(c) of the notes.
- (3): Q_8 acts on E through $SL(2, 3)$ (Lemma 3.3).
- (4): [1, Ex. 4.4].
- (P): H fixes the odd $N = 1$ current.
- (R): V_0 and V_0' are positive-definite even-lattice VOAs.

2 Set-up: twisted modules as modules over an algebra object

2.1 Parity adjunction and the bosonic base

Let $G \subset \mathrm{Aut}(V)$ be finite and solvable, acting evenly, with $\sigma \notin G$. Put $\widetilde{G} := G \times \langle \sigma \rangle$ and $U_0 := V^{\widetilde{G}} = (V^G)_0$. By (R) and Carnahan–Miyamoto [9, Main Thm], U_0 is regular. Their hypotheses are: a simple, non-negatively graded, regular VOA with a nonsingular invariant form, and a finite solvable group. Hence $\widetilde{\mathcal{C}} := \mathrm{Rep}(U_0)$ is a modular tensor category [8] **[C]**.

Let $\mathcal{S}\widetilde{\mathcal{C}}$ be the superisation of $\widetilde{\mathcal{C}}$ in the sense of [5, Def. 2.11], the *enriched supercategory of graded pairs*:

- objects are pairs $W = (W^0, W^1)$ of objects of $\widetilde{\mathcal{C}}$;
- $\mathrm{Hom}_{\mathcal{SC}}(W_1, W_2) := \mathrm{Hom}_{\widetilde{\mathcal{C}}}(W_1^0 \oplus W_1^1,\, W_2^0 \oplus W_2^1)$;
- the parity of a morphism is defined through $P_W = 1 \oplus (-1)$;
- Π exchanges the two entries.

Odd morphisms are therefore genuine morphisms of $\mathcal{SC}$: this is not a Deligne product with only even maps. With Koszul signs on the tensor product of morphisms and on the braiding, $\mathcal{SC}$ is a braided tensor supercategory with right exact tensoring [5, Prop. 2.17, Thm 2.20, Prop. 2.23] **[C]**.

Every SVOA A with $U_0 \subset A_0$ that appears below (V^G, V^E, V, V') is a commutative superalgebra in $\mathcal{SC}$, with injective unit, $\frac{1}{2}\mathbb{Z}$-graded twist and no monodromy [5, Thm 3.42] **[C]**. For such A:

- $\mathrm{Rep}\, A$, the not-necessarily-local A-modules in $\mathcal{SC}$, is a monoidal supercategory under $\otimes_A$ [5, Thm 2.53];
- $\mathrm{Rep}^0 A$ is braided [5, Thm 2.55];
- $\mathrm{Rep}^0 A$ is the category of A-modules, with the Huang–Lepowsky–Zhang structure equal to the Kirillov–Ostrik one [5, Thm 3.65], refining [7, Thm 3.4] and [6, Thm 3.14];
- induction is a tensor functor [5, Thm 2.59].

2.2 Gaiotto–Johnson-Freyd's category

In the proof of [1, Thm 4.2], the fusion product of twisted modules is provided by regularity of V^G. The bosonic case is said to be the main result of [2], and the fermionic one is obtained by replacing categories by supercategories. In [2, §4]:

- a g-twisted module is *defined* as an object X of $\mathrm{Rep}\, A$, with $A = V$ in $\mathrm{Rep}(V^G)$, satisfying $\mu_X \circ R^2 = \mu_X \circ (g^{-1} \otimes 1)$ [2, Def. 4.1];
- the product is $\otimes_A$;
- the anomaly cocycle is the associator of $\mathrm{Rep}\, A$ on the simple objects [2, eq. (5.3)].

Accordingly we *define*

$$\mathcal{D}_G := \mathrm{Rep}^0_{\mathcal{SC}}(V^G), \qquad \mathcal{F}(V, G) := \mathrm{Rep}_{\mathcal{D}_G}(V),$$

graded by Kirillov's monodromy condition, with product $\otimes_V$ over $\mathcal{D}_G$ and the associator of [5, Thm 2.52]. This reading of [1] is the one interpretive step of the paper; see Remark 7.2.

Theorem 2.1 (Identification with McRae's crossed category)**.** *Let $\mathrm{Rep}_{\mathcal{SC}}(V)$ be the braided $\widetilde{G}$-crossed supercategory of [3, Thm 4.15], which applies since $\widetilde{G} \ni \sigma$. There is a canonical equivalence of G-graded monoidal supercategories*

$$\Theta : \mathcal{F}(V, G) \xrightarrow{\simeq} \bigoplus_{g \in G} \mathrm{Rep}_g(V) \subset \mathrm{Rep}_{\mathcal{SC}}(V).$$

*It is the identity on underlying objects of $\mathcal{SC}$ and on Hom superspaces. It commutes with Π, preserves degrees, and has an even tensorator under which associators and unit isomorphisms coincide. "Canonical" means that Θ is unique up to unique monoidal isomorphism once cokernel objects (such as $V \otimes_U X$) are chosen. For fixed choices it is bijective on objects; we never need such choices. Moreover, for $G' \subset G$ the category $\mathcal{F}(V, G')$, built over $\mathrm{Rep}(V^{G'})$, is canonically equivalent, as a G'-graded monoidal supercategory, to the G'-graded part of $\mathcal{F}(V, G)$. **[P, from [5, 3] and Galois]**; proof in §7.*

From [3, Thm 4.15] **[C]**: every indecomposable object of $\mathrm{Rep}_{\mathcal{SC}}(V)$ is $\tilde{g}$-twisted for some $\tilde{g} \in \widetilde{G}$, and $\mathrm{Rep}_{\mathcal{SC}}(V)$ is braided $\widetilde{G}$-crossed; the latter is [3, Thm 2.10], which needs no parity hypothesis.

By [1, Thm 4.2] and [2, Thms 4.3, 4.7], each component $\mathcal{F}_g$ of $\mathcal{F}(V,G)$ has exactly one simple object X_g up to isomorphism and parity, and $\mathcal{F}_1 \simeq \mathrm{SVec}$.

2.3 The three layers

The class $\alpha(V,G) \in \widetilde{SH}^3(BG)$ [1, Thm 4.2] **[C]** has three layers. The dictionary is standard [1, 13].

- *Majorana layer* $m \in H^1(G;\mathbb{Z}_2)$: $m(g) = 1$ iff $\mathrm{End}(X_g) \cong \mathrm{Cl}(1)$.
- *Gu–Wen layer* $n \in C^2(G;\mathbb{Z}_2)$, defined when $m = 0$: $X_g \otimes X_h \cong \Pi^{n(g,h)} X_{gh}$.
- *Bosonic layer* $\omega \in C^3(G;U(1))$: the associator on the X_g, with $d\omega = (-1)^{n \cup n}$.

Two G-graded super-fusion categories with pointed components and the same class are equivalent by a graded equivalence that is the identity on degrees. We never use explicit cocycles. All constructions below are functors that preserve End-algebras, parities of fusion isomorphisms, and associators.

2.4 Pullback and inflation

For a homomorphism $f : G' \to G$ and a G-graded category $\mathcal{G}$, the pullback $f^*\mathcal{G}$ is defined as follows.

- $(f^*\mathcal{G})_{g'} := \mathcal{G}_{f(g')}$, with objects written $X^{(g')}$.
- Hom spaces are those of $\mathcal{G}$ within one degree, and zero between different degrees.
- $X^{(a)} \otimes Y^{(b)} := (X \otimes Y)^{(ab)}$, with the associator of $\mathcal{G}$.

Its class is $f^*\mathrm{class}(\mathcal{G})$ **[P]**. By Mac Lane strictification we may assume that $\mathcal{G}$ is strict; then $f^*\mathcal{G}$ is strict. In particular the objects $1^{(e)}$, $e \in \ker f$, have identity associators with everything.

Proposition 2.2 (Inflation)**.** *Under hypothesis (2), let $\mathcal{G} := \mathcal{F}(V,Q)$. There is an H-graded equivalence $\Phi : \pi^*\mathcal{G} \xrightarrow{\simeq} \mathcal{F}(V,H)$ that is the identity on degrees.* ***[P]***

Proof. First, $\mathcal{F}(V,Q)$ is canonically equivalent to the $s(Q)$-graded part of $\mathcal{F}(V,H)$ (Theorem 2.1). Second, $\mathrm{class}(\pi^*\mathcal{G}) = \pi^* s^* \alpha(V,H) = \pi^*\alpha = \alpha(V,H)$. Equal classes give the graded equivalence by §2.3. □

Proposition 2.3 (Coprime splitting)**.** *Let $H = E \rtimes Q$ with $\gcd(|E|,|Q|) = 1$ and $|E|$ odd. Then*

$$(s^*, \mathrm{res}_E) : \ \widetilde{SH}^3(BH) \xrightarrow{\cong} \widetilde{SH}^3(BQ) \oplus H^3(E;U(1))^Q$$

is an isomorphism. In particular, if E is non-anomalous then $\alpha(V,H) = \pi^*\alpha(V,Q)$. ***[P, given the natural filtration of $\widetilde{SH}^3$, [1, Def. 4.1]]***

Proof. The filtration. $\widetilde{SH}^3(BG) = [BG, B^3\mathrm{SAlg}^\times_{\mathbb{C}}]$ is the degree-3 group of the cohomology theory whose spectrum has homotopy groups $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{C}^\times$ in three consecutive degrees, with k-invariant Sq^2 [1, Def. 4.1]. In degree 3 it agrees with the Anderson dual of spin bordism (notes, §S2.3). It carries a three-step filtration, natural in G, whose associated graded pieces are subquotients $E_\infty^{1,\cdot}$, $E_\infty^{2,\cdot}$, $E_\infty^{3,\cdot}$ of E_2-terms. These are $H^1(G;\mathbb{Z}_2)$ (Majorana), $H^2(G;\mathbb{Z}_2)$ (Gu–Wen) and $H^3(G;U(1))$ (bosonic), and the differentials are natural. Since all groups are finite, we argue prime by prime.

Primes p dividing $|Q|$. The coefficient groups are trivial H-modules and $|E|$ is prime to p. So Lyndon–Hochschild–Serre gives $\pi^* : H^i(Q;M)_{(p)} \xrightarrow{\cong} H^i(H;M)_{(p)}$, with inverse s^*, for $M = \mathbb{Z}_2, U(1)$. Hence π^* is an isomorphism of p-localised spectral sequences from E_2 onward, and by the five lemma on the finite filtration it is an isomorphism $\widetilde{SH}^3(BQ)_{(p)} \cong \widetilde{SH}^3(BH)_{(p)}$, with inverse s^*.

Primes p dividing $|E|$. Here p is odd, so the $\mathbb{Z}_2$-layers vanish after localisation, and p-local differentials into or out of the bosonic layer are zero. Hence $\widetilde{SH}^3(BH)_{(p)} = H^3(H;U(1))_{(p)}$. Coprimality gives $H^a(Q;H^b(E;U(1))) = 0$ for $a > 0$ and $H^b(Q;U(1))_{(p)} = 0$ for $b > 0$, so $H^3(H;U(1))_{(p)} \cong H^3(E;U(1))^Q_{(p)}$ via res_E.

Conclusion. The two families of primes are disjoint, so the map is an isomorphism. Both $\alpha(V,H)$ and $\pi^*\alpha(V,Q)$ map to $(\alpha(V,Q),0)$. Restriction to subgroups is compatible with α by Theorem 2.1 (base change). □

Remark 2.4. S6.3(c) of the notes argues instead through the degree-three inclusions $H^3(X;\mathbb{C}^\times) \hookrightarrow r\widetilde{SH}^3(X) \hookrightarrow \widetilde{SH}^3(X)$, which [1] state just before Thm 4.2: the connecting maps are stable operations, which vanish in low degrees. That argument is valid. The p-local comparison above is an alternative direct proof in the coprime setting, and it gives an isomorphism rather than an injection.

3 The algebra A_ψ and Q-invariant cocycles

Work in $\pi^*\mathcal{G}$ with $\mathcal{G} = \mathcal{F}(V,Q)$ strict, and put $1_e := 1^{(e)}$ for $e \in E$. Then $1_a \otimes 1_b = 1_{ab}$, and all associators among the 1_e are identities.

Lemma 3.1. *For a normalised cochain $\psi : E \times E \to U(1)$, the object $A_\psi := \bigoplus_e 1_e$ with multiplication $\psi(a,b)\,\mathrm{id} : 1_a \otimes 1_b \to 1_{ab}$ is an associative unital algebra iff $d\psi = 1$. It is then haploid and separable, and rescaling the 1_e changes ψ by a coboundary. Every algebra structure on $\bigoplus_e X_e$ with X_e simple of degree e, $X_1 = 1$, and all multiplication components isomorphisms, is isomorphic to some A_ψ. **[P]***

Proof. With trivial associators, associativity reads $\psi(a,b)\psi(ab,c) = \psi(b,c)\psi(a,bc)$. For the last claim, write $X_e \cong \Pi^{p(e)}1_e$. The multiplication components are even isomorphisms, so $p : E \to \mathbb{Z}_2$ is a homomorphism. Since $|E|$ is odd, $p = 0$. □

Lemma 3.2 (Invariant representatives)**.** *Let Q act on E with $\gcd(|Q|,|E|) = 1$, and let $[\psi] \in H^2(E;U(1))^Q$. Then $[\psi]$ has a representative $\bar\psi$ with $\bar\psi^q = \bar\psi$ as functions, for all $q \in Q$. Two such representatives differ by $d\lambda$ with λ Q-invariant. **[P]***

Proof. $N := \prod_q \psi^q$ is invariant as a function and represents $|Q|[\psi]$. Choose n with $n|Q| \equiv 1 \pmod{|E|}$ and put $\bar\psi := N^n$. For the second claim, let $\psi' = \psi\, d\lambda$. Then $c_q := \lambda^q/\lambda \in \widehat{E}$ is a 1-cocycle. Since $H^1(Q;\widehat{E}) = 0$, we have $c_q = \mu^q/\mu$, and replacing λ by λ/μ makes it invariant. □

Lemma 3.3 (The case $E = \mathbb{F}_3^2$, $Q_8 \subset SL(2,3)$)**.** *Let $\zeta = e^{2\pi i/3}$ and $\psi_k(a,b) = \zeta^{k\det(a,b)}$ for $k \in \mathbb{Z}_3$.*

- *The ψ_k represent the three classes of $H^2(\mathbb{F}_3^2;U(1)) \cong \mathbb{Z}_3$.*
- *$g \in GL(2,3)$ sends ψ_k to $\psi_{\det(g)k}$ on the nose; in particular $SL(2,3)$ fixes each ψ_k.*
- *The Gaiotto–Johnson-Freyd Q_8 acts on every block tetracode with determinant 1.*

***[P]**; exhaustive check in `psi_cocycle_check.py`.*

Proof. Each ψ_k is bimultiplicative. The commutator pairing $\zeta^{2k\det}$ separates the three classes, and $\det(ga,gb) = \det(g)\det(a,b)$. The determinant statement was verified on the certificate (notes, §9.2). □

The conjugation action. Fix $q \in Q$ and a simple $X = X_q \in \mathcal{G}_q$. By (4), $\operatorname{End} X = \mathbb{C}$, so X is invertible, with isomorphisms $\mathrm{ev} : X^* \otimes X \to 1$ and $\mathrm{coev} : 1 \to X \otimes X^*$. Put $\bar{X} := X^{(s(q))}$ and $\bar{X}^* := (X^*)^{(s(q)^{-1})}$. (The unit cannot be placed in degree $s(q)$ for $q \neq 1$.)

Lemma 3.4. *The functor $c_q(Y) := \bar{X} \otimes Y \otimes \bar{X}^*$, with tensorator $\mathrm{id} \otimes \mathrm{ev} \otimes \mathrm{id}$, is a monoidal autoequivalence covering $h \mapsto s(q)hs(q)^{-1}$. After the identifications $u := \mathrm{coev}^{-1} : c_q(1_e) \to 1_{qe}$, it is* strictly *monoidal on the objects 1_e. Consequently $c_q(A_\psi) = A_{\psi^q}$ on the nose.*

If Φ is as in Proposition 2.2, the VOA transport $T_{s(q)} : M \mapsto M \circ s(q)^{-1}$ satisfies $T_{s(q)} \circ \Phi \cong \Phi \circ c_q$ on the E-part. ***[P; the second claim uses the crossed braiding of [3, Thms 2.10, 4.15]]***

Proof. Strictness. Both $u \circ J_{1_a,1_b}$ and $u \otimes u$ are maps $XX^*XX^* \to 1$ in $\mathcal{G}$, independent of a and b. Precomposing with the isomorphism $\mathrm{coev} \otimes \mathrm{coev}$ gives 1 for both, by the zigzag identity. Also $u \circ \mathrm{coev} = \mathrm{id}$. Hence c_q is strictly monoidal on the 1_e, and transporting the multiplication gives $\psi'({}^q a, {}^q b) = \psi(a, b)$.

Comparison with $T_{s(q)}$. For an invertible $s(q)$-twisted M, the crossed braiding $c_{M,-} : M \otimes N \to T_{s(q)}(N) \otimes M$ is natural and monoidal in N (crossed hexagon). So $T_{s(q)} \cong M \otimes - \otimes M^*$ as monoidal functors. Taking $M = \Phi(\bar{X})$ gives the claim. □

Hence the Q-action on orbifold algebra objects is $\psi \mapsto \psi^q$, and hypothesis (3) is intrinsic. From now on ψ is a Q-invariant cocycle (Lemma 3.2) and $A := A_\psi$.

4 Bimodules: the categorical core

Let $\mathrm{Bimod}_A := \mathrm{Bimod}_A(\pi^*\mathcal{G})$. It is Q-graded by support on cosets qE. For $X \in \mathcal{G}_q$ define the bimodule $B_X := A \otimes X^{(s(q))} = \bigoplus_e X^{(eq)}$, with

- left action of $1_{e'}$: multiplication by $\psi(e', e)$ on $X^{(eq)}$;
- right action of $1_{e'}$: multiplication by $\psi(e, {}^q e')$ on $X^{(eq)}$.

For $\chi \in \widehat{E}$, the bimodule B_X^χ is B_X with its right action multiplied by $\chi(e')$.

Proposition 4.1. *(a) B_X and B_X^χ are bimodules. Right associativity holds for all e, a, b iff $\psi({}^q a, {}^q b) = \psi(a, b)$, which is why invariance on the nose is needed.*

(b) $\mathrm{Ind} : X \mapsto B_X$, $f \mapsto \bigoplus_e f^{(eq)}$, *is fully faithful on Hom superspaces. In particular* $\operatorname{End} B_{X_q} = \operatorname{End} X_q$.

(c) Ind *is monoidal: $B_X \otimes_A B_Y \cong B_{X \otimes Y}$ via the free-module isomorphism. On homogeneous summands this composite carries the balancing factor $\psi(e, {}^q f)$. The two threefold coefficients agree,*

$$\psi(e, {}^q f)\, \psi(e\, {}^q f, {}^{qr} g) = \psi(f, {}^r g)\, \psi(e, {}^q (f\, {}^r g)),$$

by the cocycle identity and invariance. So no new 3-cocycle on Q arises, and the associator of the B's is that of $\mathcal{G}$.

(d) Consequently $\mathrm{Ind} : \mathcal{G} \xrightarrow{\simeq} \mathrm{Ind}(\mathcal{G}) \subset \mathrm{Bimod}_A$ *is an equivalence of Q-graded super-fusion categories onto its image. It preserves m, n and ω, so* $\operatorname{class}(\mathrm{Ind}\, \mathcal{G}) = \alpha$.

[P]

Proof. (a) Left associativity, and the commutation of the left and right actions, are instances of the cocycle identity. For right associativity, apply the cocycle identity at $(e, {}^q a, {}^q b)$; this reduces it to $\psi({}^q a, {}^q b) = \psi(a, b)$.

(b) Supports on distinct cosets are disjoint. A bimodule map from the free left module B_X is determined by its restriction to $X^{(q)}$, which lands in degree $s(q)$. The only possibly odd morphism is f, and it is never moved past another odd morphism, so no Koszul sign arises.

(c) The coequaliser defining $\otimes_A$ splits on free modules, and s is a homomorphism. Associativity constraints are induced from those of $\pi^*\mathcal{G}$, which are those of $\mathcal{G}$. □

Lemma 4.2 (Classification of bimodules)**.** *Every simple A-bimodule in $\pi^*\mathcal{G}$ is, up to parity, isomorphic to exactly one $B^{\chi}_{X_q}$, with $q \in Q$ and $\chi \in \widehat{E}$. Hence there are exactly $|H|$ simple bimodules up to parity, each with* End $= \mathbb{C}$*.* ***[P]***

Proof. 1. Every left A-module is a sum of free modules $A \otimes X_q^{(q)}$, since $A \otimes 1_e \cong A$. Moreover $\mathrm{Hom}_A(A \otimes X_q, A \otimes X_{q'}) = \delta_{qq'}\mathbb{C}$, using $m = 0$.
2. A simple bimodule P is supported on a single coset, so as a left module $P \cong (A \otimes X_q) \otimes W$ for a super vector space W.
3. The right action of $1_{e'}$, composed with the inverse of the canonical right action of B_{X_q}, is an even left-module automorphism $\rho(e') \in GL(W)$. Right associativity of P and of B_{X_q} makes ρ a representation of E.
4. Since E is abelian, W splits into characters, so $P \cong \Pi^p B^{\chi}_{X_q}$.
5. Comparing right actions on $\mathrm{Ind}(f)$ shows $B^{\chi} \cong B^{\chi'}$ iff $\chi = \chi'$.

□

Remark 4.3. The $|E|$ bimodules $B^{\chi}_{X_q}$ on each coset correct the earlier sketch, which claimed B_{X_q} was the only one. The extra ones are the sectors twisted by the dual symmetry $\widehat{E}$.

5 The gauged theory and the bimodule dictionary

Inputs.

- *(Gal) Quantum Galois correspondence.* For a simple SVOA W and finite $G \subset \mathrm{Aut}\, W$, $K \mapsto W^K$ is a bijection from subgroups of G to subalgebras containing W^G. This is [10, Thm 6.2], attributed there to Dong–Mason and Hanaki–Miyamoto–Tambara; the super Schur–Weyl decomposition it rests on is [4, Thm 3.2] [**C**°].
- *(Ext) Extensions and orbifolds.*
  - SVOA extensions $L' \supset B = V^E$ correspond to commutative superalgebras in $\mathcal{D}$ [5, Thm 3.42], [7, Thm 3.2], and isomorphisms over B correspond to algebra isomorphisms.
  - *Definition.* $V//_{\psi}E$ is the extension L' for which $\mathrm{Ind}_B^L(L') = L \otimes_B L' \cong \Phi(A_\psi)$ as algebras in Rep(L). This is Kirillov's correspondence. Under the equivariantisation equivalence of [3, Thm 4.17], applied to $\widetilde{E} = E \times \langle\sigma\rangle$, which contains parity, and composed with the forgetful functor, the induction Ind_B^L is the functor that forgets the E-equivariant structure. So an orbifold is an E-equivariant algebra whose underlying algebra is A_ψ. This motivates the definition but is not used in the proofs. Existence for each $[\psi]$ is [1, Thm 4.2]; in the application the extensions are explicit lattice SVOAs.
  - Automorphisms of V normalising E transport extensions; on algebra objects the transport is given by Lemma 3.4.

Proposition 5.1 (Theorem 1.1(i))**.** *Let $K := \{g \in \mathrm{Aut}\, V' : g(V^E) = V^E,\ g|_{V^E} \in H|_{V^E}\}$. Then $1 \to \widehat{E} \to K \xrightarrow{\bar{\pi}} Q \to 1$ is exact and split, its complements are $\widehat{E}$-conjugate, and $(V')^K = V^H$.* ***[P from (Gal), (Ext)]***

Proof. Let $\chi \in \widehat{E}$ act on the twisted-sector summand N_e of V' by $\chi(e)$; then $(V')^{\widehat{E}} = V^E$. For $q \in Q$, the extension $s(q) \cdot V'$ (which is V' with B acting through $s(q)^{-1}$) has induced algebra $T_{s(q)}\Phi(A_\psi) \cong \Phi(A_{\psi^q})$ by Lemma 3.4. By (3) and Lemma 3.1 this is isomorphic to $\Phi(A_\psi)$. By

(Ext), the resulting isomorphism of extensions composes with the transport to give $\phi_q \in \operatorname{Aut} V'$ with $\phi_q|_{V^E} = s(q)|_{V^E}$. By (Gal), $H|_{V^E} \cong H/E \cong Q$ and $\ker(K \to Q) = \operatorname{Aut}(V'/V^E) = \widehat{E}$. Schur–Zassenhaus applies since $\gcd(|\widehat{E}|, |Q|) = 1$, and $\widehat{E}$ is abelian. Finally $(V')^K = ((V')^{\widehat{E}})^Q = V^H$. □

Put $\widetilde{H} = H \times \langle\sigma\rangle$ and $\widetilde{K} = K \times \langle\sigma'\rangle$. Here $\sigma' \notin K$: otherwise $h\sigma$ would fix V^E for some $h \in H$, so $h\sigma \in E$ by (Gal), and then $\sigma \in H$. Both groups contain parity and have the common bosonic fixed algebra $U_0 = V^{\widetilde{H}} = V'^{\widetilde{K}}$. Set $B := V^E$, $L := V$ and $L' := V'$ in $\mathcal{SC}^{\sim}$, and $\mathcal{D} := \operatorname{Rep}^0_{\mathcal{SC}^{\sim}}(B)$. Decompose $L = \bigoplus_\chi V_\chi$ under E, and $L' = \bigoplus_e N_e$ under $\widehat{E}$.

(Inv) The V_χ and N_e are invertible objects of $\mathcal{D}$.

- In general, this follows from [10, Thm 6.3, Prop. 4.17] under their twisted-module hypotheses **[C]**.
- In the application, $B = V_{\Lambda_E}$ with $\Lambda_E = \Lambda(C_i \cap U_b^\perp)$, and the V_χ, N_e are lattice coset modules $V_{\lambda+\Lambda_E}$. These are simple currents by the lattice fusion rules [14] **[C°]**.

Lemma 5.2 (Charge–flux monodromy)**.** *In $\mathcal{D}$, $c_{N_e,V_\chi} c_{V_\chi,N_e} = \chi(e)^{-1}$. Here the label e of N is* defined *as the degree of the invertible L-module $M_e := L \otimes_B N$. **[P]***

Proof. By [2, Def. 4.1] and [3, Def. 2.3], $\mu \circ M_{L,M_e} = \mu \circ (e^{-1} \otimes 1)$. Restrict to $V_\chi \otimes (B \otimes_B N_e)$ and use modules in stages. There μ is the canonical isomorphism onto $V_\chi \otimes_B N_e$, which is nonzero. □

Put $A := L \otimes_B L' = \bigoplus_N \operatorname{Ind}(N) \in \operatorname{Rep}_{\mathcal{D}}(L)$, the sum running over the $|E|$ summands of L'. By (Ext), $A \cong \Phi(A_\psi)$ has exactly one simple summand in each degree $e \in E$. Hence $N \mapsto e(N)$ is a bijection onto E; write N_e for the summand with label e, so that $A = \bigoplus_e M_e$ with $M_e = \operatorname{Ind}(N_e)$. (*Consistency check:* the multiplication components $M_e \otimes_L M_{e'} \to M_{ee'}$ are Ind of the multiplication $N_e \otimes_B N_{e'} \to N_{ee'}$ of V'. These are nonzero because V' is simple and its vertex operators do not vanish on nonzero vectors [3, Lemma 4.18], and Ind is faithful.)

Theorem 5.3 (Bimodule dictionary)**.** *The functor*

$$\Psi : \mathcal{F}(V', K) \overset{2.1}{\cong} \bigoplus_{k \in K} \operatorname{Rep}_k(L') \subset \operatorname{Rep}_{\mathcal{SC}^{\sim}}(L') \simeq \operatorname{Rep}_{\operatorname{Rep} B}(L') \longrightarrow \operatorname{Bimod}_A(\mathcal{F}(V, H)), \qquad \Psi(N) = L \otimes_B N,$$

*is an equivalence of super-fusion categories. It commutes with Π, and it is compatible with cosets: $\Psi(\mathcal{F}(V', K)_k)$ is supported on $s(\bar{\pi}k)E$. **[P]***

Proof. Monoidal. We use three facts.

- *Modules in stages.* $\operatorname{Rep}_{\mathcal{SC}^{\sim}}(L) \simeq \operatorname{Rep}_{\operatorname{Rep} B}(L)$ as monoidal supercategories, by the remark in [11, §3.6]. Both relative tensor products are the same quotient of $X \otimes Y$, and the coequalisers split by separability.
- *Central algebras.* For a commutative algebra R in the centre of $\operatorname{Rep} B$, $\operatorname{Rep} R$ is monoidal and the free functor is monoidal [12], [5, Thm 2.59].
- *Transport of bimodules.* Additive monoidal functors preserve split coequalisers.

Then Ψ is the composite of three monoidal functors,

$$\operatorname{Rep}(L') \subset \operatorname{Bimod}_{L'}(\operatorname{Rep} B) \to \operatorname{Bimod}_{\operatorname{Ind} L'}(\operatorname{Rep} L).$$

The right action uses the half-braiding, with one braiding convention fixed throughout.

Cosets. Monodromies multiply, and L is local over B, so B-types are preserved.

Fully faithful (central vectors). Let $P = \Psi(N)$. Bimodule maps $A \to P$ are the central vectors $x : 1 \to P$. Commuting with L is automatic. Decompose $x = \sum_\chi x_\chi$ with $x_\chi : 1 \to V_\chi \otimes_B N$. By the hexagon, naturality and Lemma 5.2, the right action of N_e on x_χ equals $\chi(e)$ times the left action. No Koszul sign appears: the only possibly odd morphism is x_χ, and naturality pairs it with id_{N_e}. The left action $\ell_e(\mathrm{id} \otimes x_\chi)$ vanishes only if $x_\chi = 0$, by invertibility of the N_e-multiplication. So $x_\chi = 0$ for $\chi \neq 1$, and the central vectors are exactly $\mathrm{Hom}(L', N)$. Full faithfulness follows by rigidity, since Ψ is monoidal.

Essentially surjective.

- By Theorem 2.1 and Lemma 4.2, Bimod_A restricted to H-degrees has $|H|$ simples up to parity, each with End $= \mathbb{C}$.
- $\mathcal{F}(V', K)$ is pointed with $|K| = |H|$ simples [1, Thm 4.2].
- Full faithfulness and Π-compatibility map simples injectively, up to parity, onto simples.
- It follows also that $m(V', K) = 0$.

Restriction. Degrees in $\widetilde{K} \setminus K$ go to Ramond-type cosets, outside $\mathcal{F}(V, H)$. □

6 Proof of Theorem 1.1(ii) and independence of choices

Proof of (ii).

1. Let $\mathcal{D}' := \Phi(\mathrm{Ind}\,\mathcal{G}) \subset \mathrm{Bimod}_{\Phi(A)}(\mathcal{F}(V, H))$. By Proposition 4.1, $\mathcal{D}'$ is a full monoidal subcategory with class α.
2. Under Ψ^{-1}, its $|Q|$ simples go to simples in $|Q|$ distinct degrees of $\mathcal{F}(V', K)$. These degrees form a subgroup $\widetilde{Q}$, which maps isomorphically to Q by coset compatibility, so $\widetilde{Q}$ is a complement of $\widehat{E}$.
3. Since each degree carries a single simple, $\Psi^{-1}(\mathcal{D}')$ is the $\widetilde{Q}$-graded part of $\mathcal{F}(V', K)$. This is canonically equivalent to $\mathcal{F}(V', \widetilde{Q})$ as a graded monoidal supercategory (Theorem 2.1, base change). Hence $\alpha(V', \widetilde{Q}) = \alpha$.
4. Any complement is $Q' = c\widetilde{Q}c^{-1}$ with $c \in \widehat{E}$. Conjugation by c is a graded equivalence compatible with $\bar{\pi}$, so $\alpha(V', Q') = \alpha$.

□

Proposition 6.1 (Independence)**.** *The conclusion is independent of three choices:*

- *the complement $s(Q)$, since all complements are E-conjugate and $s'^*\pi^* = \mathrm{id}$, and conjugate complements give bimodules related by $1_e \subset A$;*
- *the invariant representative ψ, since another one is related by an invariant rescaling (Lemma 3.2), which is a Q-compatible algebra isomorphism;*
- *the equivalence Φ, since* class$(\mathcal{D}')$ *is computed intrinsically.*

[P]

7 Proof of Theorem 2.1

Put $U := V^G$, so that $\mathcal{D}_G = \mathrm{Rep}^0 U$. Write $I^U_{V,X} : V \boxtimes X \to V \otimes_U X$ for the cokernel map.

Proof.

1. *Objects.* Define $\Theta(X, \mu_X) := (X, \mu_X \circ I^U_{V,X})$.
   - Associativity transfers in both directions, because the associator of $\mathrm{Rep}\,U$ is characterised through I^U [5, Thm 2.52].
   - By [3, Thm 4.15(1)], $\Theta(X) = \bigoplus Y_{\tilde{g}}$ with $Y_{\tilde{g}}$ $\tilde{g}$-twisted. Each summand is local over U. On $U \boxtimes Y_{\tilde{g}}$ this gives $\nu \circ ((\tilde{g}^{-1}|_U - 1) \boxtimes 1) = 0$, so every $u' = \tilde{g}^{-1}u - u \in U$ acts by zero on $Y_{\tilde{g}}$. By the non-vanishing lemma [3, Lemma 4.18] (V simple, $\tilde{G} \ni \sigma$), this forces $u' = 0$, so $\tilde{g}|_U = \mathrm{id}$. Then $\tilde{g} \in G$ by (Gal) and (P).

- Conversely, the action ν of a g-twisted Y with $g \in G$ is U-balanced, by associativity together with the commutativity of V [5, Thm 3.42(3)]. So ν factors uniquely through I^U.

2. *Morphisms, parity and* Π. Morphisms on both sides are $\mathcal{S}\widetilde{\mathcal{C}}$-morphisms commuting with the action, and I^U is epi. So Θ is the identity on Hom superspaces, and Π is the same on both sides.
3. *Degrees.* We have $I^U \circ R^2 = (R^U)^2 \circ I^U$ [5, eq. (2.28)], and g fixes U. So Kirillov's and McRae's monodromy conditions coincide.
4. *Cokernels are preserved.* The quotients below are taken in three categories, $\mathcal{D}_G = \mathrm{Rep}^0 U \subset \mathrm{Rep}\, U$ and $\mathcal{S}\widetilde{\mathcal{C}}$. They agree for three reasons.
   - *Rep U to SC.* For an even (or parity-homogeneous) morphism in $\mathrm{Rep}\, U$, the cokernel computed in $\mathcal{S}\widetilde{\mathcal{C}}$ carries a unique U-action. This uses right exactness of $U \boxtimes -$ [5, Prop. 2.23], and the result is the cokernel in $\mathrm{Rep}\, U$ [5, Prop. 2.32].
   - *Local modules to Rep U.* The locality condition $\mu \circ M_{U,-} = \mu$ passes to quotients by naturality of M and surjectivity of the quotient map. So $\mathrm{Rep}^0 U$ is closed under these cokernels, and cokernels in $\mathrm{Rep}^0 U$ are cokernels in $\mathrm{Rep}\, U$ [5, Thm 2.55, Prop. 2.56].
   - *All three split.* $\widetilde{\mathcal{C}}$ is semisimple because U_0 is regular, and U and V are separable. So each coequaliser defining a relative tensor product is split by a separability idempotent, and split coequalisers are preserved by every additive functor.

   In particular the forgetful functors $\mathrm{Rep}^0 U \to \mathrm{Rep}\, U \to \mathcal{S}\widetilde{\mathcal{C}}$ preserve the cokernels used in the next step. Without this, a cokernel internal to the local-module category need not be a cokernel in the ambient category.
5. *Tensorator.* $I^{V/\mathcal{D}}_{X,Y} \circ I^U_{X,Y}$ and $I^{V/\mathcal{S}\widetilde{\mathcal{C}}}_{X,Y}$ are cokernels of the same pair $X \boxtimes V \boxtimes Y \rightrightarrows X \boxtimes Y$. A map coequalising the V-pair coequalises the U-pair, and $\boxtimes$ is right exact [5, Prop. 2.23]. This gives a unique even natural isomorphism τ compatible with the V-actions [5, Props. 2.38, 2.39, 2.46, 2.47].
6. *Associators and units.* [5, Thm 2.52] characterises each associator as the unique morphism compatible, through the cokernel maps, with the associator of the base. Composing the characterisations for $V/\mathcal{D}$ and $U/\mathcal{S}\widetilde{\mathcal{C}}$, both associators satisfy the same equation with respect to the same epimorphism, so they are equal. The units are treated the same way using [5, Props. 2.40, 2.42].

Base change. Replace U by $B' = V^{\widetilde{G'}}$ for $G' \subset G$; the same steps apply. Combining this with the theorem for G and for G' gives a canonical graded monoidal equivalence between $\mathcal{F}(V, G')$ and the G'-graded part of $\mathcal{F}(V, G)$. □

Remark 7.1 (McRae's Remark 4.16)**.** [3, Rem. 4.16] leaves open whether McRae's twisted intertwining operators agree with Huang's general definition for non-commuting g_1, g_2; they agree with Xu's for commuting ones. Theorem 2.1 compares two *categorical* constructions, so that question does not arise here.

Remark 7.2 (The interpretive step)**.** We read the tensor structure in [1, Thm 4.2] as Kirillov's $\otimes_V$, because that is the construction in the source the proof cites. If one instead *defines* the anomaly through Huang's analytic tensor product of twisted modules, the comparison of Remark 7.1 becomes necessary, and Q_8 is non-abelian. In that case the accurate status is: *the categorical argument proves the lemma under the stated hypotheses and the graded monoidal identification; completing that identification closes the application to the GJF anomaly.*

8 Application: the Q_8 neighbour chain

For each of the 23 steps $C_i \to C_{i+1} = (C_i \cap U_b^\perp) + U_b$ of the notes (§S6.2), set $V = V_{\Lambda(C_i)}$, $E \cong \mathbb{Z}_3^2$ and $Q = Q_8$. That $V' = V_{\Lambda(C_{i+1})}$ is an orbifold in the sense of (Ext), with a Q_8-invariant class, takes three separate steps.

(A1) *Cosets and degrees (lattice intersection).* The summands of V' over $B = V_{\Lambda_E}$, with $\Lambda_E = \Lambda(C_i) \cap \Lambda(C_{i+1}) = \Lambda(C_i \cap U_b^\perp)$, are $N_\nu = V_{\nu+\Lambda_E}$ for $\nu \in \Lambda(C_{i+1})/\Lambda_E$. Each induces the L-module $V_{\nu+\Lambda(C_i)}$, whose degree is the image of ν in $(\Lambda(C_i) + \Lambda(C_{i+1}))/\Lambda(C_i) \cong E$. This map is injective because the intersection is Λ_E, and it is bijective because both sides have order 9. So each degree occurs exactly once in $\mathrm{Ind}(V')$.

(A2) *The multiplication cocycle.* The components $M_e \otimes_L M_{e'} \to M_{ee'}$ of $\mathrm{Ind}(V')$ are Ind of the vertex operators $N_\nu \otimes N_{\nu'} \to N_{\nu+\nu'}$. These are nonzero, since $Y(e^\nu, z)e^{\nu'} = \epsilon(\nu,\nu')z^{\langle\nu,\nu'\rangle}e^{\nu+\nu'} + \dots$ with $\epsilon \neq 0$. Nonzero maps between invertible objects are isomorphisms, so Lemma 3.1 gives $\mathrm{Ind}(V') \cong A_{\psi'}$ for a well-defined class $[\psi'] \in H^2(E; U(1)) \cong \mathbb{Z}_3$. Its value is not needed below. The count in the notes (§9.2) shows that V' is one of the three orbifolds $V//_{\psi_k}E$, that is, $[\psi']$ is one of the classes of Lemma 3.3.

(A3) *Invariance.* By Lemma 3.3, Q_8 acts on E through $SL(2,3)$ and fixes each ψ_k on the nose. So hypothesis (3) holds for $[\psi']$, and Lemma 3.2 supplies the invariant representative used in §4 (here ψ_k itself).

Theorem 1.1 gives $\alpha(V', Q') = \alpha(V, Q_8)$ for every complement Q'. Every complement is a homomorphic lift of the monomial $Q_8 \subset O(\Lambda(C_{i+1}))$, since $\widehat{E}$ acts trivially on the Heisenberg currents. So it is conjugate to the Gaiotto–Johnson-Freyd lift by §S6.3(e). Hence

$$\alpha(V_{\Lambda(C)}, Q_8) = \alpha(V_{\mathbb{Z}^{96}}, Q_8) = 24\,\alpha(\mathrm{Fer}(8), Q_8) = 0.$$

The inputs *not* supplied by Theorem 1.1 are kept separate:

1. the exact computations: the frame code, the neighbour chain, and Lemma 3.3;
2. the lift identification, §S6.3(e), via $H^2(Q_8, \Lambda) = 0$;
3. existence of the extensions, which are explicit lattice SVOAs;
4. additivity of anomalies under $\otimes$;
5. $\Omega_3^{\mathrm{Spin}}(BQ_8) \cong \mathbb{Z}_8 \oplus \mathbb{Z}_4^2$, which has exponent 8.

9 Status

Step	Status
Inflation, algebra A_ψ, invariant cocycles, conjugation (§§2–3)	[**P**], given GJF Thm 4.2 classification [**C**]
Bimodule embedding and classification (§4)	[**P**]
K, Schur–Zassenhaus (Prop. 5.1)	[**P**], from (Gal) [**C**°] and (Ext) [**C**]
Bimodule dictionary (Thm 5.3)	[**P**], from CKM [**C**], McRae Thm 4.15 [**C**], (Inv)
(Inv)	application: lattice fusion [**C**°]; general: DJX Thm 6.3 [**C**] under its hypotheses
Identification with GJF (Thm 2.1)	[**P**] from CKM/McRae, on the reading of Rem. 7.2
Regularity (R) for V_0'	application: lattice VOA; general: Lin Thms 3.10/3.11 (ABD, HKL) [**C**°] plus the CFT-type proviso

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